



Investment Society
Imperial College London

Lecture 3

Equities

Securities and Education Certificate

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)

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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

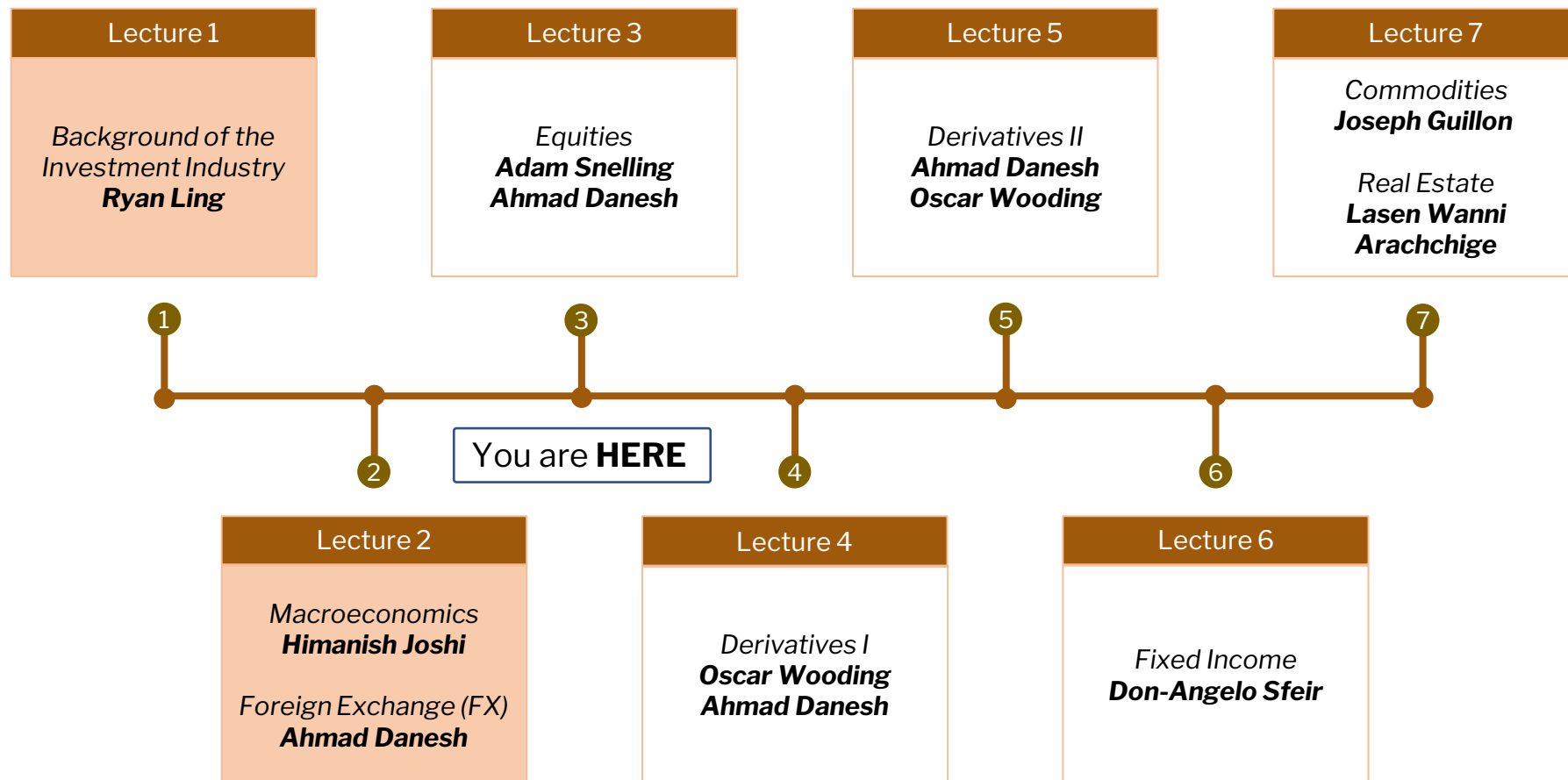
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Let's begin...



Ahmad Danesh
Lead Lecturer
Founder, QT Capital

Lecture 3

Accounting Intuition

Accounting

What does accounting aim to do?

👉 **MISCONCEPTION** – Accounting concerns cash flow and profitability.

👉 **REALITY** – Accounting measures the operations of a business.

👉 **AIM** – Break down a business's operations with numerical data.

👉 **HOW** – The three key financial statements.

Balance Sheet

Income Statement

Cash Flow Statement

Your notes

Cash-based accounting

Why might measuring operations through cash be a bad idea?

Consider a bakery e.g. *QT bakery* with a fiscal year (FY) from January 2022 – January 2023.

An order for £1,000 of product is received on October 29th 2022 but will be paid in March 2023.

Ingredients are purchased on October 29th 2022 but will be paid in April 2023.

Transaction	Value (£)	Transaction Date	Payment Date
Sale of Goods	1000	29 th October, 2022	1 st March, 2023
Purchase of Inventory	500	29 th October, 2022	1 st April, 2023

Your notes

Cash-based accounting

What would the profitability of the business look like for 2022?

Assume this order is the only order QT bakery received.

Writing out sales and costs for when **money was received** i.e. cash-based accounting:

For the FY Jan 2022 – Jan 2023 <i>Cash-Based</i>	
Account	Value (£)
Sales	0
Costs	0
Profit	0

Your notes

Cash-based accounting

Cash-based accounting doesn't account for the **operations** of the business.

👉 Measuring cash does **NOT** reflect a business's operations.

👉 Services and goods can be sold *before* cash is received.

For the FY Jan 2022 – Jan 2023 Cash-Based		
Account	Value (£)	
Sales	0	×
Costs	0	×
Profit	0	×

Your notes

Accrual-based accounting

Why not try accounting for a business based on **WHEN** an activity is completed?

Assume this order is the only order QT bakery received.

We recognise when the products have been sold, **NOT** when the money has been delivered.

For the FY Jan 2022 – Jan 2023 Accrual-Based		
Account	Value (£)	
Sales	1000	✓
Costs	500	✓
Profit	500	✓

Your notes

Accrual-based accounting

So, accrual-based accounting works! How do we account for cash though?

👉 Use an accounts receivables (**AR**) account for ordered products and unpaid cash.

👉 Use the **cash flow statement** to show exact movements of cash.

For the FY Jan 2022 – Jan 2023 Accrual-Based

Account	Value (£)	
Sales	1000	✓
Costs	500	✓
Profit	500	✓

Your notes

How the three financial statements link together

So, now we know why accounting works, how do the financial statements work?

👉 The **balance sheet** shows a snapshot of a business's assets and liabilities.

👉 The **income statements** links two consecutive balance sheets via net income.

👉 The **cash-flow statement** highlights the cash reserve changes between two balance sheets.

Knowledge of financial statements is **CRITICAL** for some sectors of an Investment Bank...

Investment Banking

Equity Research

Anyone interested in a SEC team-led workshop on accounting fundamentals?

Your notes

Keep an eye out later this lecture for more news!

Basic Equities' Technicals

Let's solve some of these common misconceptions.



Justin Wong Credit Suisse Intern

Accounting and Valuation Workshop
11th – 13th October, 2022

Some frequently asked questions concerned...

EBIT vs EBITDA

Enterprise Value

The WACC

Your notes

So, let's answer some of them this lecture!

The Income Statement

Where does the EBIT and the EBITDA play? The income statement.

Consider this mock income statement below.

Income Statement (£000s)	
Revenue	1000
Cost of Goods Sold (COGS)	(800)
GROSS PROFIT	200
SG&A	(90)
Depreciation	(10)
OPERATING PROFIT	100
Interest	(10)
INCOME BEFORE TAXES	90
Tax (30%)	(27)
NET PROFIT	63

The Income Statement

Where does the EBIT and the EBITDA play? The income statement.

☞ EBITDA means **E**arnings **B**efore **I**nterest, **T**axes, **D**epreciation and **A**mortisation.

Income Statement (£000s)		
Revenue	1000	
Cost of Goods Sold (COGS)	(800)	
GROSS PROFIT	200	
SG&A	(90)	110
Depreciation	(10)	+ 10
OPERATING PROFIT	100	
Interest	(10)	+ 10
INCOME BEFORE TAXES	90	
Tax (30%)	(27)	+ 27
NET PROFIT	63	63

☞ **Net profit** is £63,000.

☞ Go from the bottom and work up.

☞ **EBITDA**
 = 63 + 27 + 10 + 10 + 0
 = 110
 i.e. £110,000

☞ **EBITDA**
 = Net income (profit)
 + Interest
 + taxes
 + Depreciation + Amortisation

The Income Statement

We've done EBITDA. What about EBIT?

☞ EBIT means **E**arnings **B**efore **I**nterest and **T**axes **ONLY**.

Income Statement (£000s)	
Revenue	1000
Cost of Goods Sold (COGS)	(800)
GROSS PROFIT	200
SG&A	(90)
Depreciation	(10)
OPERATING PROFIT	100
Interest	(10)
INCOME BEFORE TAXES	90
Tax (30%)	(27)
NET PROFIT	63

↑ 100

+ 10

+ 27

● 63

☞ **Net profit** is £63,000.

☞ Go from the bottom and work up.

☞ **EBIT**
 = 63 + 27 + 10
 = 100
 i.e. £100,000

☞ **EBIT**
 = Net income (profit)
 + Interest
 + taxes
 = **OPERATING PROFIT!**

What does EBIT and EBITDA tell you?

Never use numbers without knowing what they mean!

👉 Operating Profit / EBIT tells you the raw income generated by a business's **MAIN** operations.

👉 EBITDA further reverses the depreciation and amortisation deductions.

👉 Depreciation/Amortisation do **NOT** represent actual loss of cash reserves.

👉 Depreciation/Amortisation spreads out the purchase cost of an asset over its lifetime.

👉 **TAKE-HOME MESSAGE:** EBIT/EBITDA measure income from a business's **MAIN** operations.

Your notes

Enterprise Value

So, what does Enterprise Value (EV) actually do? An example should help.

👉 Businesses can raise cash in either two ways.

Sell shares
i.e. issue **equity**

Take out a loan
i.e. issue **debt**

Total **cash** generated = total value of **equity** issued (i.e. market capitalisation) + total **debt** issued.

If a new business issues £1 million in shares and loans £1 million, it has £2 million **TOTAL** in cash.

👉 Idle cash (i.e. not being used) does not contribute to the business's operations (i.e. enterprise).

The **enterprise value** of the new business is £0 → no operations are taking place.

Enterprise Value (EV) = Market Capitalisation + Total Debt – Cash.

Enterprise Value

So, what does Enterprise Value actually do?

👉 Enterprise Value (EV) measures the **value** of a business by factoring its operational spend.

Enterprise Value (EV) = Market Capitalisation + Total Debt – Cash.

👉 EV also measures the amount of money to buy out an entire company.

The business's cash reserves helps recoup part of the purchase value you pay.



Pay the entire value of Debt and Equity on the right side (**RHS**) to buy the left side (**LHS**)

Your notes

Market Capitalisation

What does it **actually** measure?

👉 Market capitalisation is the total value the market gives to the business's **equity**.

Market Capitalisation = Share price × No. of issued shares

👉 A business's accountants may overvalue the total value of shareholders' equity.



👉 Markets may undervalue the business.

'Market Cap' measures the value of **equity**.

Your notes

The WACC

Oh my... This is going to be a huge turning point in financial teaching.

👉 The WACC means the **W**eighted **A**verage **C**ost of **C**apital

👉 You need to understand **W**eighted **A**verages.

We are going to intuitively derive the WACC from first principles.

What is the mean of the following set of numbers? {1, 2, 3, 4, 5}

$$\frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

Your notes

Weighted Averages

Great! GCSE Maths at play there. Why do we care?

☞ An alternative method of calculation involves multiplying each number in the set by $\frac{1}{5}$

$$\frac{1 + 2 + 3 + 4 + 5}{5} = \frac{1}{5}(1) + \frac{1}{5}(2) + \frac{1}{5}(3) + \frac{1}{5}(4) + \frac{1}{5}(5) = 3$$

☞ Each 'element' in the set is **weighted** by a factor of $\frac{1}{5}$

☞ This is a **weighted average** calculation of the mean.

The weights were identical, with all numbers weighted at $\frac{1}{5}$ each to sum to 1.

Your notes

Weighted Averages

What if the weights do not sum to 1?

👉 We can divide by the *sum* of the weights.

If three exams are weighted by percentage at 25%, 25% and 50%, the total weight is 100(%).

For exam results of 30/60, 40/60 and 50/60, the **weighted average** mark is **NOT** 40/60!

$$\frac{25(30) + 25(40) + 50(50)}{100} = 0.25(30) + 0.25(40) + 0.5(50) = 42.5$$

👉 Decimal-based weights offer faster computation.

Your notes

The WACC

What does this have to do with the WACC?

☞ Recall what ways money i.e. capital is raised.

Sell shares
i.e. issue **equity**

Take out a loan
i.e. issue **debt**

☞ Both **equity** and **debt** have a cost associated with them (C_E and C_D respectively) → **cost of capital**.

If a firm's cost of capital is 25% equity and 75% debt, the cost of capital must be a **weighted** average.

$$\frac{(25 \times C_E) + (75 \times C_D)}{100} = WACC$$

☞ The **WACC** is the **W**eighted **A**verage of the **C**ost of **C**apital!

Your notes

The WACC

Extend the case to a general algebraic for a general situation.

☞ Consider the total **value** of equity and debt to be **E** and **D** respectively.

$$\left(\frac{E}{E+D} \times C_E\right) + \left(\frac{D}{E+D} \times C_D\right) = WACC$$

☞ Define **V** = E + D

$$\left(\frac{E}{V} \times C_E\right) + \left(\frac{D}{V} \times C_D\right) = WACC$$

☞ C_D & C_E are percentages → C_D is the interest rate, C_E is expected rate of return from shareholders.

☞ Relabel C_E as R_d

$$\left(\frac{E}{V} \times R_d\right) + \left(\frac{D}{V} \times R_e\right) = WACC$$

☞ Relabel C_D as R_e

Your notes

The WACC

Let's extend the discussion to include the tax rate.

☞ **Debt** is useful as it can reduce the amount of tax paid.

Income Statement (£000s)	
Revenue	1000
Cost of Goods Sold (COGS)	(800)
GROSS PROFIT	200
SG&A	(90)
Depreciation	(10)
OPERATING PROFIT	100
Interest	(10)
INCOME BEFORE TAXES	90
Tax (30%)	(27)
NET PROFIT	63

☞ If no debt existed, the business would pay **£30,000** in tax.

☞ The £10,000 interest payment saved **£3,000** in tax payment.

☞ Interest is a 'tax-deductible' expense.

☞ The cost of debt is the interest paid per year – any tax saved.

☞ Here, the cost of debt is $£10,000 - £3,000 = £7,000$

☞ This can also be calculated as $£10,000 \times (1 - 0.3) = £7,000$

The WACC

The cost of debt must also include the **tax shielding** effect that interest provides.

☞ Factoring in the **‘tax shielding’** (last slide), we achieve the final form of the WACC formula.

$$\left(\frac{E}{V} \times R_e\right) + \left(\frac{D}{V} \times R_d\right)(1 - T) = WACC$$

☞ T is the tax rate.

☞ The WACC weights the cost of equity R_e and cost of debt R_d based off their proportions in the firm.

☞ The WACC is often used as a discount rate for DCF calculations.

An understanding of the derivation and intuition is **HIGHLY** advised!

Your notes

BONUS MATERIAL A Capital Asset Pricing Model (CAPM) discussion is available in the extended notes!

Let's resume...



Adam Snelling

Lecturer

Senior Research Analyst, Imperial Blockchain Group
Cryptocurrency Analyst, QT Capital (2021-22)

Lecture 3

Equities

Asset Classes

What is an asset?

👉 An **asset** is anything of value.

👉 More specifically in this context, anything of **economic** value.

👉 Asset classes are groups of assets that behave in a similar manner.

Asset Classes

Fixed Income
(e.g. Bonds)

Equities

Commodities

Cryptocurrencies

Your notes

Asset Classes

What is an asset?

👉 There are many assets. Below, we see examples of a few.

Asset Classes

Fixed Income (e.g.
Bonds)

Equities

Commodities

Cryptocurrencies

Low Risk

Medium Risk

High Risk

High Risk

👉 Each asset class is characterised by similar properties e.g. the risk level.

Your notes

What is Equity?

Now that we know what an asset is, we look at equity in particular.

👉 **Equity** represents ownership of a company.

👉 If you buy equity in a company, you are a part-owner.

👉 How much of the company you own depends on how many shares you own.

Equities = Assets - Liabilities

👉 The value of a company's equity is what is leftover once all assets are sold and all liabilities are paid.

Your notes

Types of Equity

Now we know what equity is, what are the different types?

👉 There are two main types of equity.

Types of Equity

Private Equity

Companies **NOT** listed on a stock exchange

Shares must be purchased privately from shareholders.

Public Equity

Companies listed on a stock exchange

Shares can be purchased through a stock broker.

Your notes

Stock Exchange

Before we delve deeper, we need to cover stock exchanges.

Consider buying shares of a company, ABC.

A 'supply chain' exists for retail investors to be able to purchase the shares:

Inform your stock broker to buy 10 shares of ABC.

→ Your stock broker goes to the exchange and buys 10 ABC shares.

→ Your stock broker returns to your account to sell you 10 ABC shares.

What would this look like if you wanted to sell shares?

How does the stock broker make profit in this situation?

👉 The purpose of a stock exchange is to provide **liquidity**.

Private Equity

What do we mean by private equity?

👉 **Private equity** refers to ownership in companies not listed on a stock exchange.

👉 All companies start out as private firms.

👉 In private companies, share ownership is largely concentrated between founders & angel investors.

👉 A private equity firm invests in both public **AND** private companies, to sell later at a profit.

Private equity firms pool capital from institutional investors.

Wealthy
Individuals

Pension Funds

They acquire firms with high growth potential with the aim to make them more efficient and profitable.

They aim to then sell their shares for a profit or through an IPO (Initial Public Offering).

Public Equity

Ok. If that's private, then what is public?

👉 Companies go public by an **IPO (Initial Public Offering)** i.e. shares are made public for the first time.

👉 There are many reasons why a company might decide to do an IPO:

Raise capital
to invest for further growth

Pay off
existing debt it owes

👉 Anyone can invest in a public company via the 'supply chain'.

👉 **'Capital'** is the assets required to produce goods or services e.g. cash or cash equivalents.

Your notes

Company Metrics

How do we measure company performance?

☞ There are many metrics and ratios used to compare companies and measure their performance.

☞ Many of these can provide a quick guess as to how a company performs in its sector.

☞ **It is important to look at these metrics alongside other data.**

☞ Some of the most popular metrics used by investors are:

PE Ratio

PS Ratio

EBITDA

EV

Your notes

P/E Ratio

What is the P/E ratio?

👉 P/E ratio stands for '**P**rice to **E**arnings' ratio.

👉 As the name suggests, it compares the price of a companies shares to its earnings.

P/E Ratio = Share Price ÷ Earnings per Share

👉 Share Price refers to the current market price of one share of the company.

Earnings per Share = Earnings ÷ No. of shares outstanding

👉 P/E ratio can show if a company is overvalued or undervalued compared to its profits.

P/E Ratio

How might we calculate the P/E ratio?

Consider calculating the P/E ratio of ABC.

Share Price = £20

Net Income = £1 million

Shares Outstanding = 100,000

Earnings Per Share (**EPS**) = £1 million ÷ 100,000 = £10

P/E Ratio = £20 ÷ £10 = 2

👉 P/E ratios are useful for comparing companies in the same sector, so investors find the best deal.

👉 P/E is **NOT** foolproof – it has flaws with some types of companies.

Young Companies
with no earnings

How do you think we might do this?

P/S Ratio

What is a P/S ratio?

👉 P/S ratio stands for a company's **P**rice to **S**ales ratio.

👉 It is similar to PE ratio **BUT** does not take into account a firm's expenses.

P/S ratio = Share Price ÷ Sales per share

👉 This can be very useful when trying to value companies that are not currently making any profit.

👉 A **forward** P/S ratio uses *forecasted* sales based on our analysis, **NOT** *current* sales.

Your notes

P/S Ratio

Let's look at how we would calculate this.

Consider calculating the P/S ratio of ABC.

Share Price = £20

Revenue (last Q) = £2.5 million

Shares Outstanding = 100,000

Sales Per Share (**SPS**) = $\text{£2.5 million} \div 100,000 = \text{£25}$

P/S Ratio = $\text{£20} \div \text{£25} = 0.8$

 **NOTE:** P/S ratio is always less than P/E ratio by definition.

 The P/S ratio ignores the business's expenses, which is crucial to the viability of a business.

This won't only apply to Tesla.

👉 EV stands for the enterprise value of a company.

👉 This metric tries to provide a wholistic view of what the current value of a company is.

EV = market capitalization + debt – cash or cash equivalents

Market cap

This is the 'market value' of the company: **No. of shares × the share price.**

Debt

The amount of money the company owes.

Cash or cash
equivalents

Cash equivalents are assets that can be easily turned into cash (i.e. *liquid*).

Let's have a go at this one.

Consider calculating the EV of ABC.

Share Price = £20

Shares Outstanding = 100,000

Debt = £500,000

Cash (Equivalents) = £200,000

Market Cap. = $£20 \times 100,000 = £2 \text{ million}$

$EV = £2 \text{ million} + £500,000 - £200,000 = £2.3 \text{ million}$

👉 Think of Enterprise Value as the amount of money needed for someone to buy a company.

👉 EV can be combined into ratios with other data points to provide other valuable metrics .

Can you think of any?

EBITDA

What on earth is that?

👉 EBITDA stands for **e**arnings **b**efore **i**nterest, **t**ax, **d**epreciation and **a**mortization.

👉 More accurate to measure profit → it removes the effects of "exterior" factors.

EBITDA = earnings + interest + taxes + depreciation + amortization

Interest

These are interest payments the company needs to make on debt it owes.

Depreciation

Refers to the loss of value of tangible (physical) assets, e.g. machinery.

Amortization

Similar to depreciation, it is the loss of value of intangible assets e.g. a patent on company software.

EBITDA

Let's try this one.

Consider calculating the EBITDA ratio of ABC using the **income statement**

Net Income = £1 million Interest = £25,000 Taxes = £200,000 Depreciation & Amortisation = £50,000

$\text{EBITDA} = \text{£1 million} + \text{£25,000} + \text{£200,000} + \text{£50,000} = \text{£1.275 million}$

👉 EBITDA ignores costs associated with financing decisions, focusing on the core business model.

👉 EBITDA is rejected in asset-intensive industries, where depreciation is huge on a company's profits.

Your notes

EV/EBITDA vs P/E

What should you choose?

👉 **P/E** and **P/S** ratios are used for comparing companies usually in the same industry.

👉 This means a **P/E** or **P/S** value on its own is essentially meaningless.

👉 **EV** and **EBITDA** are alternative ways of determining the value of a company.

👉 So, comparing the **EVs** of two companies is also useless.

👉 We can define a new ratio **EV/EBITDA** (sometimes used instead of **P/E**)

👉 **EV** is more accurate for the price of a company, and **EBITDA** is a better tool for its earnings.

Time Value of Money

There is even more value to money!? Great!

👉 Over time, money loses its value.

Inflation

Opportunity Cost

👉 In other words, £100 **NOW** is worth more than £100 in the future because of the time difference.

Suppose I gave you the option of £100 now, or £105 in a year from now.

What would you choose?

What do you need to consider?

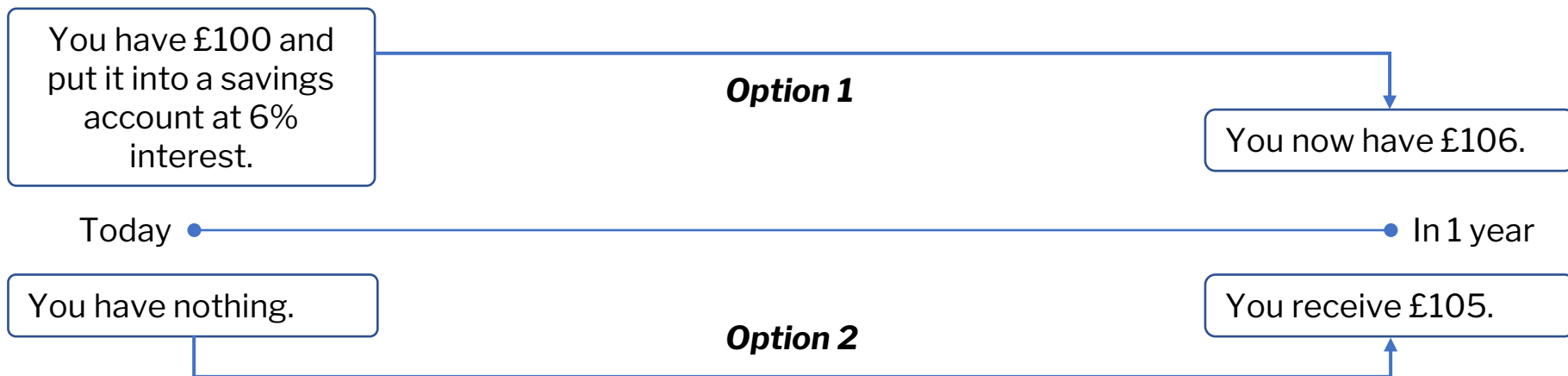
Your notes

Time Value of Money

There is even more value to money!? Great!

Take the £100 now and invest that into an asset that will give me greater than 5% return in a year.

Suppose, I can put my money in a savings account with 6% annual interest.



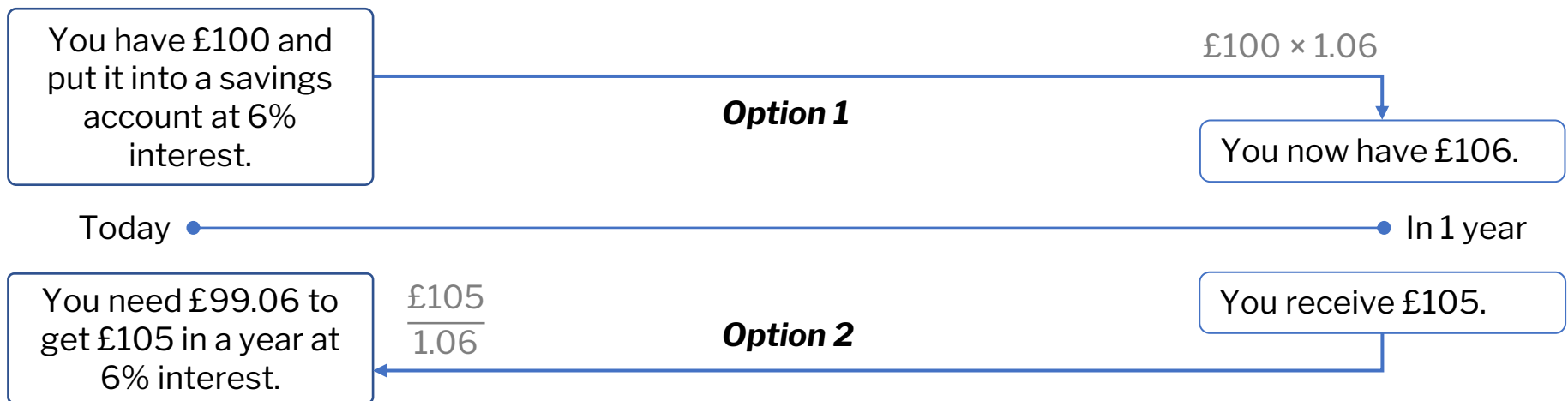
Your notes

Time Value of Money

What if we look in reverse?

Now, suppose you want £105 in the future and need to calculate how much you need to invest *now*.

You divide £105 by the interest rate (in this case, 1.06) to obtain how much you need to invest *now*.



Your notes

Time Value of Money

Phew! Let's recap that.

☞ For an asset e.g. a savings account with 6% interest, £100 now is worth **more** than £105 in a year.

☞ In reverse, we calculated the **present values (PV)** of both options.

☞ You discount the **future value (FV)** to find out how much you pay now for that return in the future.

☞ You use a **discount rate** e.g. the interest rate.

☞ In finance, you discount forecasts of future value using your rate of return to find the present value.

☞ We used this method for just one year of **cash flow** → We can extend this idea to multiple years.

Discounted Cash Flow

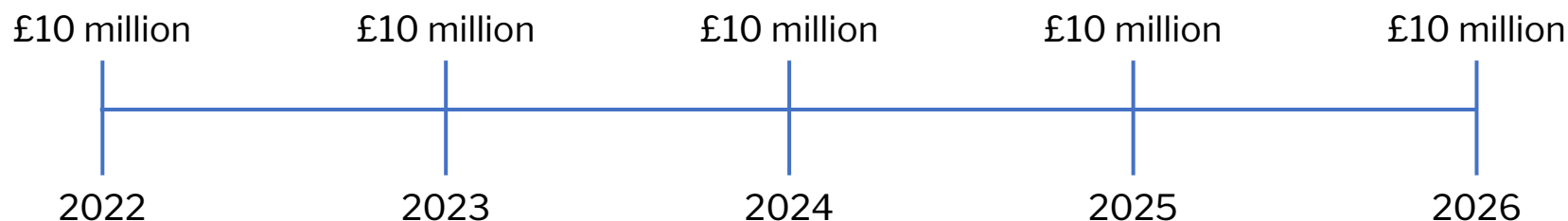
That leads us on nicely to the discounted cash flow method.

👉 **Discounted Cash Flow (DCF)** analysis is a method of valuing a company.

👉 It uses ideas relating to the time value of money as a discounted rate of return.

👉 It presents the idea of using projected cash flows to value a company currently.

Suppose ABC Ltd. makes £10 million net profit every year. For a return p.a. of **10%**, how much do you pay to buy ABC?



👉 It is **NOT** $5 \times £10\text{m} = £50\text{m}$! Money **now** is worth more than the same amount of money in 2026.

Discounted Cash Flow

How do we go about calculating a price (in PV) for ABC?

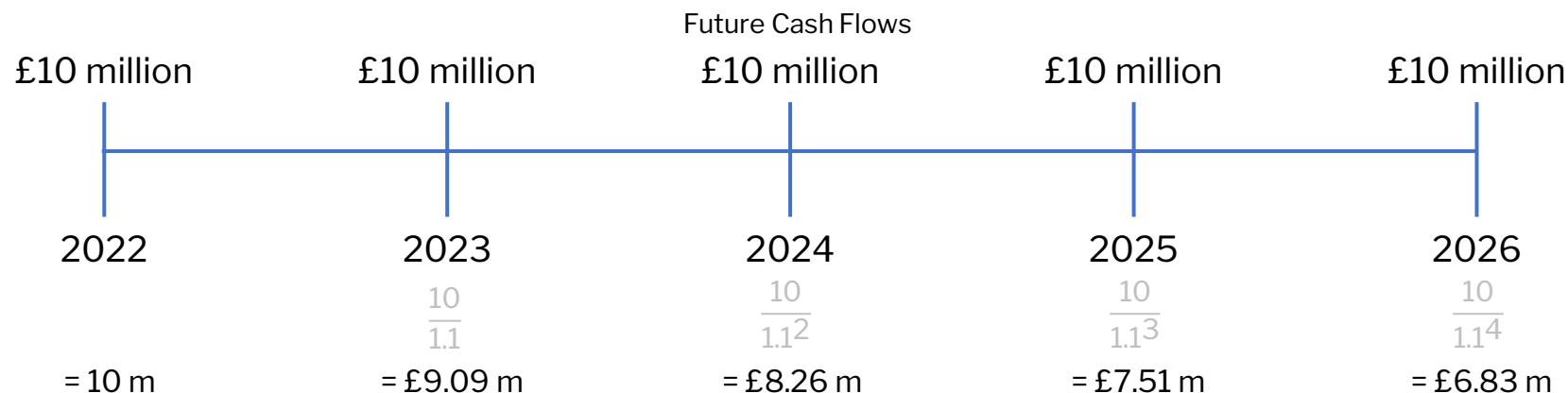
☞ Discount the cash flows to its present value using our rate of return of 10%:

Present value for 2023 cash flow = $£10 \text{ m} \div 1.1 = \text{£}9.09 \text{ m}$

☞ Our discount rate (rate of return) is on an annual basis → For 2024, *discount* the cash flow twice.

Present value for 2024 cash flow = $£10 \text{ m} \div 1.1^2 = \text{£}8.26 \text{ m}$

☞ We can continue doing this for every year...



Present Value (of Future Cash Flows)

Discounted Cash Flow

Discounted Cash Flow

So, how do we calculate the final **total** present value?

☞ The total sum of our present values is:

$$\begin{array}{rclclcl} \text{Sum of present values} & = & \text{£10 m} & + & \text{£10m}/1.1 & + & \text{£10m}/(1.1^2) & + & \text{£10m}/(1.1^3) & + & \text{£10m}/(1.1^4) \\ & = & \text{£10 m} & + & \text{£9.09 m} & + & \text{£8.26 m} & + & \text{£7.51 m} & + & \text{£6.83 m} \\ & = & \text{£41.7 m} & & & & & & & & \end{array}$$

☞ We should pay **£41.7 million** now for £50 million in future cash flows.

Present Value = $\frac{\text{Cash Flow 1}}{(1+r)} + \frac{\text{Cash Flow 2}}{(1+r)^2} + \dots + \frac{\text{Cash Flow } n}{(1+r)^n}$ for discounting n years of cash flow.

☞ Here, the discount rate r was 10% (or 0.1).

Other discount rates include e.g. the WACC
Depending on analysis **type**

Your notes

Discounted Cash Flow

We could also use a little shortcut.

☞ Some of you may have noticed that our sum looks like a geometric series.

☞ Recall the formula for a geometric series:

$$S_n = \frac{a(1-r^n)}{(1-r)}, \quad \text{where } n = \text{number of terms, } a = \text{first term and } r = \text{common ratio}$$

☞ Using this formula, we see the present value is:

$$\text{Present Value} = \frac{10 \left(1 - \frac{1}{1.1^5} \right)}{1 - \frac{1}{1.1}} = \text{£}41.7 \text{ m}$$

☞ *The same answer.*

☞ **NOTE** the cash flows don't have to be constant → they can change depending on your analysis.

See the notes for further information!

Macroeconomic Environment

The greater economy has an effect as well.

☞ Countries go through periods of high growth and contraction in a cyclical pattern.

☞ This cyclical pattern is known as an **economic cycle**.

☞ The value of equities broadly follows the country's economic cycle.

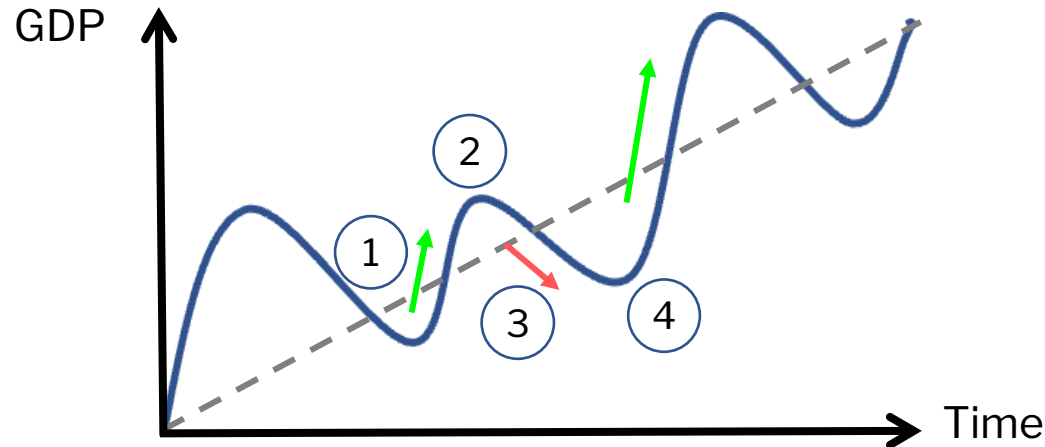
☞ Recall that asset classes are groups of assets that behave in a similar manner.

☞ **The total value of a stock market in a country broadly follows the country's economic cycle.**

Your notes

Economic Cycle

What are the different stages?



👉 **1. Growth Stage:** high level of output, increased productivity and low levels of unemployment.

👉 **2. Peak/slowdown stage:** economic growth slow down, resulting in high levels of inflation.

👉 **3. Contraction stage:** economic output decreases, more difficult for businesses to survive.

👉 **4. Trough/recovery stage:** GDP becomes stable usually due to government policy.
The economy is ready to start growing again.

How to profit from Equities

How do people make money from equities?

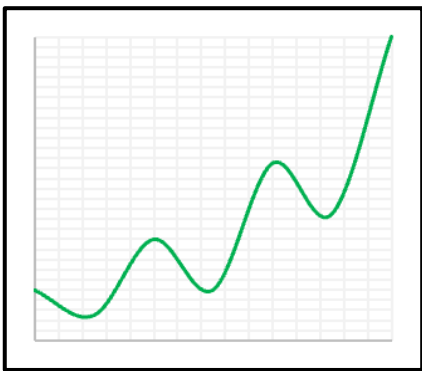
☞ Equities are seen as 'risky' assets → Asset value correlates to the company performance.

☞ The *potential* return for equities is **higher** than less risky assets, such as *bonds*.

Why do people invest in equities i.e. buy stocks?

Growth Stocks

Equities *may* become more valuable in the future



Dividend Stocks

Equities may give high dividend payouts



Source: Reuters

How to profit from Equities

If you've bought some equity, how do you make money from it?

👉 **Growth stocks** are seen as 'risky' but can provide reward in the long term.

Companies during **Infancy**

Companies producing
sector-specialised goods

Companies in
emerging economies

Can you think of a sector which is likely to have a lot of growth stocks?

👉 **Dividend stocks** are seen as less 'risky' **BUT** depends on the *dividend yield*.

Dividend yield = Annual dividends per share ÷ Share price

👉 Dividend stocks are usually more mature companies with little growth potential.

Fund

This last section is really fun(d). Sorry.

👉 A fund is a collection of assets.

👉 They provide a low-cost opportunity to invest in a diverse range of assets all at once.

👉 Equity funds provide a collection of different company's shares in one investment vehicle.

👉 **There are many different types of funds for different types of investors.**

Index Fund

Active Fund

ETF

Mutual Fund

Your notes



NOTE There will be overlap between some of these funds.

Index Fund

Aaah... passive investing. Sounds easy to me.

👉 An **index fund (passive fund)** tracks the value of the underlying assets in the fund.

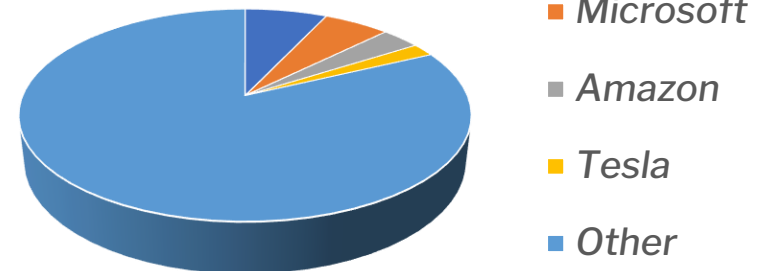
👉 e.g. the *S&P 500* → tracks the top 500 companies in the US, weighted by market cap.

👉 An index fund allows you to invest in all the index's assets. The fund's value tracks the index's value.

👉 *Index funds are usually low-cost*
It's just an algorithm following a number.

👉 1 share of an S&P 500 index fund:
 7% of 1 Apple share
 5.5% of 1 Microsoft share, etc.

S&P 500 Allocation



Actively-managed Fund

What's the opposite to a passive fund?

👉 An **actively-managed fund** is the opposite to an index fund.

👉 An active fund *handpicks* certain stocks to go in the fund → **NO** market value tracking.

👉 A **fund manager** chooses which stocks go in the fund

👉 The chosen stocks relate to the fund's *goals*.

Long-term returns

ESG

👉 Dependency on fund manager → Active funds tend to be more *risky* than index funds

👉 You pay for the fund manager's expertise → active funds have higher costs.

Exchange-Traded Fund

Now, how do I buy into these funds?

👉 An **exchange-traded fund** (ETF) refers to how a fund is traded.

👉 ETFs are traded on the stock exchanged just like shares in a public company.

👉 They can be traded throughout the day by anyone.

👉 They are subject to the effects of supply and demand, just like stocks on the stock market.

👉 **Index ETF**: a fund that tracks an index comprised of different companies, which is publicly traded.

Your notes

Mutual Fund

Mutual? We might have a lot in common then.

☞ A mutual fund cannot be traded on a stock exchange throughout the day → *opposite* of ETFs.

☞ Mutual fund shares must be purchased usually from the fund's creator **OR** are traded once per day.

☞ The rules for mutual funds are *usually* stricter than ETFs.

What do you think these could be?

☞ Sometimes, there is a minimum amount you have to invest.

☞ **NOTE** most mutual funds are actively-managed, so they have a fund manager.
BUT the goal of the mutual fund might just be to track the returns of an index.

Fund Summary

So, what do we know?

👉 An **index fund** is one that just tracks the value of an index.

👉 An **exchange-traded fund (ETF)** can be traded throughout the trading day on the stock market.

👉 An **active fund** is one that is managed by a fund manager.

👉 A **mutual fund** is a fund that cannot be traded on a stock exchange.
It can only be traded once per day, or directly through the provider.

Index **ETF**

Index **Mutual Fund**

Active **ETF**

Active **Mutual Fund**

VUSA – Vanguard fund that tracks the value of the S&P 500.

Emerging Markets Stock Index Fund – Vanguard fund that tracks performance of the MSCI.

JREG – J.P. Morgan Global Research Enhanced Index Equity UCITS ETF.

Active U.K. Equity Fund – Vanguard fund comprising of primarily U.K. shares.